

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12400842
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Yang; Wenbing et al.

Method of cleaning chamber components with metal etch residues

Abstract

A method of cleaning residue containing ruthenium (Ru) residue on at least one surface of a component of a semiconductor processing chamber is provided. The residue is exposed to a Ru cleaning composition comprising at least one of hypochlorite and O.sub.3 based chemistries, wherein the Ru cleaning composition removes the Ru residue.

Inventors:	Yang; Wenbing (Campbell, CA), Tan; Samantha SiamHwa (Newark, CA), Lin; Ran (Fremont, CA), Mukherjee; Tamal (Fremont, CA), Zhou; Chunhong (Fremont, CA), Kang; Xiaoyu (Union City, CA), Pan; Yang (Los Altos, CA), Shih; Hong (Cupertino, CA)
Applicant:	Lam Research Corporation (Fremont, CA)
Family ID:	1000008778270
Assignee:	Lam Research Corporation (Fremont, CA)
Appl. No.:	18/258926
Filed (or PCT Filed):	December 06, 2021
PCT No.:	PCT/US2021/062044
PCT Pub. No.:	WO2022/159183
PCT Pub. Date:	July 28, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240304428 A1	Sep. 12, 2024

Related U.S. Application Data

Publication Classification

Int. Cl.: **H01J37/32** (20060101); **B08B3/04** (20060101); **B08B3/08** (20060101); **B08B9/08** (20060101); **C11D7/08** (20060101); **C11D7/10** (20060101); **C11D7/32** (20060101); **C11D7/50** (20060101); **C23C16/44** (20060101)

U.S. Cl.:

CPC **H01J37/32862** (20130101); **B08B3/08** (20130101); **B08B9/08** (20130101); **C11D7/08** (20130101); **C11D7/10** (20130101); **C11D7/3218** (20130101); **C11D7/5013** (20130101); **C23C16/4407** (20130101); **B08B3/04** (20130101); **H01J2237/022** (20130101)

Field of Classification Search

CPC: H01J (37/32862); C23C (16/4407); B08B (3/00-14); B08B (9/08-46); B08B (9/00-46); H01L (21/00-86)

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7107998	12/2005	Greer et al.	N/A	N/A
7546840	12/2008	Hasebe et al.	N/A	N/A
9257638	12/2015	Tan et al.	N/A	N/A
10767257	12/2019	Vitiello et al.	N/A	N/A
2002/0066466	12/2001	Tan	134/28	C23G 5/036
2003/0013314	12/2002	Ying	257/E21.311	H01L 21/32136
2004/0144320	12/2003	Hasebe	118/724	C23C 16/4404
2008/0121249	12/2007	Gatineau	134/2	B08B 7/00
2010/0218788	12/2009	Bao	134/3	C23C 16/4407
2015/0079786	12/2014	Tan et al.	N/A	N/A
2015/0280113	12/2014	Tan	438/3	C23F 1/12
2019/0185999	12/2018	Shanbhag et al.	N/A	N/A
2022/0203410	12/2021	Scherer	N/A	B08B 3/047

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2000-200782	12/1999	JP	N/A
2001-284317	12/2000	JP	N/A
2001284317	12/2000	JP	N/A
2003-013232	12/2002	JP	N/A
2005-187880	12/2004	JP	N/A
2019-504929	12/2018	JP	N/A
2020-190014	12/2019	JP	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion from International Application No.

PCT/US2021/062044 dated Apr. 7, 2022. cited by applicant

Wei Lin, Rong-Wei Zhang, Seung-Soon Jang, Ching-Ping Wong, and Jung-Il Hong. “‘Organic Aqua Regia’—Powerful Liquids for Dissolving Noble Metals.” *Angewandte Chemie International Edition*, Sep. 17, 2010. cited by applicant

Notice for Reason of Refusal from Japanese Application No. 2023-542480 dated Jun. 24, 2025 with Machine Translation. cited by applicant

Korean Office Action from Korean Application No. 10-2023-7028006 dated Jul. 24, 2025 with Machine Translation. cited by applicant

Primary Examiner: Barr; Michael E

Assistant Examiner: Zhang; Richard Z.

Attorney, Agent or Firm: Beyer Law Group LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of priority of U.S. Application No. 63/139,231, filed Jan. 19, 2021, which is incorporated herein by reference for all purposes.

BACKGROUND

(1) The disclosure relates to a method of cleaning metal residues from components of a plasma processing chamber.

(2) During semiconductor wafer processing, features may be etched through a metal containing layer. In the formation of magnetic random access memories (MRAM) or resistive random-access memory (RRAM) devices, a plurality of thin metal layers or films may be sequentially etched. For MRAM a plurality of thin metal layers may be used to form magnetic tunnel junction stacks. The thin metal layers may contain ruthenium (Ru) and other metals such as cobalt (Co), iron (Fe), palladium (Pd), nickel (Ni), boron (B), platinum (Pt), tantalum (Ta), molybdenum (Mo), titanium (Ti), manganese (Mn), magnesium (Mg), chromium (Cr), iridium (Ir), tungsten (W), copper (Cu), aluminum (Al), hafnium (Hf), indium (In), tin (Sn), gold (Au), and silver (Ag).

(3) Such metal containing layers may be etched in a plasma processing chamber. Etch metal residues may deposit on plasma facing surfaces of components of the plasma processing chamber during the plasma etching process. Too much residue changes the performance of the chamber and may also create contaminants and may also make the component unusable. A change in the performance of the component may cause nonuniformities. Contaminants may cause the failure of devices being manufactured. If the component is unusable and must be replaced, a replacement component may be expensive.

(4) The background description provided here is for the purpose of generally presenting the context of the disclosure. Information described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

SUMMARY

(5) To achieve the foregoing and in accordance with the purpose of the present disclosure, a method

of cleaning residue containing ruthenium (Ru) residue on at least one surface of a component of a semiconductor processing chamber is provided. The residue is exposed to a Ru cleaning composition comprising at least one of hypochlorite and O.sub.3 based chemistries, wherein the Ru cleaning composition removes the Ru residue.

(6) In another manifestation, a method of cleaning residue comprising a noble metal residue from a component of a semiconductor processing chamber is provided. The noble metal residue is exposed to a noble metal cleaning composition comprising thionyl chloride and pyridine, wherein the noble metal cleaning composition removes the noble metal residue.

(7) These and other features of the present disclosure will be described in more detail below in the detailed description and in conjunction with the following figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings and in which like reference numerals refer to similar elements and in which:

(2) FIG. 1 is a high level flow chart of an embodiment.

(3) FIGS. 2A-B are schematic views of a stack processed in an embodiment.

(4) FIG. 3 is a schematic view of an etch reactor that may be used for etching.

(5) FIG. 4A is an enlarged cross-sectional view of part of a component of the plasma processing chamber with a residue layer.

(6) FIG. 4B is an enlarged cross-sectional view of part of the component after the residue layer has been removed.

(7) FIG. 5 schematic view of the component in a wet bath.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(8) The present disclosure will now be described in detail with reference to a few preferred embodiments thereof as illustrated in the accompanying drawings. In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. It will be apparent, however, to one skilled in the art, that the present disclosure may be practiced without some or all of these specific details. In other instances, well known process steps and/or structures have not been described in detail in order to not unnecessarily obscure the present disclosure.

(9) During semiconductor wafer processing, features may be etched through a metal containing layer. In the formation of magnetic random access memories (MRAM) or resistive random-access memory (RRAM) devices, a plurality of thin metal layers or films may be sequentially etched. For MRAM a plurality of thin metal layers may be used to form magnetic tunnel junction stacks. The thin metal layers may contain ruthenium (Ru) and other metals such as cobalt (Co), iron (Fe), palladium (Pd), nickel (Ni), boron (B), platinum (Pt), tantalum (Ta), molybdenum (Mo), titanium (Ti), manganese (Mn), magnesium (Mg), chromium (Cr), iridium (Ir), tungsten (W), copper (Cu), aluminum (Al), hafnium (Hf), indium (In), tin (Sn), gold (Au), and silver (Ag). In various embodiments, the thin metal layers may comprise of Group III, IV, and V element metals (III-V metals).

(10) Such metal containing layers may be processed, such as etched, in a plasma processing chamber. After a metal plasma etch, the plasma chamber components are contaminated with multiple etching species including metals in both metallic and compound form, and silicon species from process wafer or mask materials. Contaminants on the chamber wall will cause severe issues in integrated circuit (IC) fabrication by affecting chamber plasma conditions and thus wafer-to-wafer repeatability. For etching most metals in an MRAM magnetic tunnel junction (MTJ), halogen

chemistry is applied to assess the etching efficacy. Metal containing residues from wafer processing are redeposited on chamber walls. X-ray photoelectron spectroscopy analysis of chamber wall surfaces reveals metals are mostly in compound form, such as metal fluoride (MF_x, M: metal). In addition, the chamber wall surface is also coated with silicon oxide layers from the process wafer, hard mask materials, or etching chemicals. A mixture of metal/metal compounds and silicon oxide form contamination layers inside the chamber. This contamination results in several issues including flaking off metal particles onto wafers and process drifts by releasing multiple atoms from the chamber wall during wafer processing. Too much residue changes the performance of the chamber and may also create contaminants and may also make the component unusable. A change in the performance of the component may cause nonuniformities. Contaminants may cause the failure of devices being manufactured. If the component is unusable and must be replaced, a replacement component may be expensive.

(11) If the residue cannot be cleaned from the component, the component must be discarded and replaced. Some of these residues are noble metal residue. Noble metals are metals that are highly resistant to chemical attack. Noble metals include Pd, Au, Ru, Pt, rhodium (Rh), osmium (Os), and iridium (Ir).

(12) Embodiments provide a method of cleaning components with metal residues of at least Ru and other noble metal residues. To facilitate understanding, FIG. 1 is a high level flow chart of a process used in an embodiment. Various embodiments may have more or less steps. In addition, in various embodiments the steps may be performed in different orders or simultaneously. Metal layers are processed in a chamber (**104**). One example chamber may be a plasma processing chamber. For example, the metal layers may form a stack on a wafer for providing an MRAM, RRAM, cobalt interconnects, etc. FIG. 2A is a schematic cross-sectional view of an example MRAM stack **200** on a process wafer that is processed in an embodiment. The stack **200** is on a substrate with a silicon or silicon oxide (Si/SiO₂) layer **204**. A first tantalum (Ta) layer **208** is over the Si/SiO₂ layer **204**. A platinum (Pt) layer **212** is over the first Ta layer **208**. A cobalt platinum alloy (CoPt) layer **216** is over the Pt layer **212**. A magnesium oxide (MgO) layer **220** is over the CoPt layer **216**. A cobalt iron boron (CoFeB) layer **224** is over the MgO layer **220**. A second Ta layer **228** is over the CoFeB layer **224**. A ruthenium (Ru) layer **232** is over the second Ta layer **228**. A patterned mask is formed over the stack **200**. In this embodiment, the patterned mask comprises a titanium nitride layer **236**, under a SiO₂ layer **240**, under a Ru layer **244**.

(13) The stack **200** may be processed so that one or more of the metal containing layers are etched. To do so, the stack may be subjected to one or more etch processes. FIG. 2B is a schematic cross-sectional view of the stack **200** after an exemplary process is completed on of the stack **200**. As shown, the processing of the stack has etched the first Ta layer **208**, the Pt layer **212**, the CoPt layer **216**, the MgO layer **220**, the CoFeB layer **224**, the second Ta layer **228**, and the Ru layer **232**. Some of the patterned mask may also be etched.

(14) As a result of the processing of the stack **200**, materials from the Ru, SiO₂, TiN, CoFeB, MgO, CoPt, Pt, and/or Ta layers may be deposited on plasma facing surfaces of the plasma processing chamber. For example, as a result of the etch, residues of tantalum, beryllium, platinum, manganese, cobalt, iron, ruthenium, magnesium, titanium, and silicon are deposited on plasma facing surfaces of the chamber. After processing, the stack may be removed. This may be done by removing the wafer from the chamber. Subsequently, another stack on a wafer may be placed in the process chamber for processing. As each stack is etched, further coating of residues will form on the plasma facing surfaces of the chamber. In some embodiments, more than a thousand wafers are processed in the chamber.

(15) In one embodiment, all processing may be performed in a single plasma etch chamber. In other embodiments, processing may be performed in different chambers. FIG. 3 schematically illustrates an example of a processing chamber system **300** that may be used in an embodiment. The processing chamber system **300** includes a plasma reactor **302** having a plasma processing chamber

304 therein. A plasma power supply **306**, tuned by a power matching network **308**, supplies power to a transformer coupled plasma (TCP) coil **310** located near a dielectric inductive power window **312** to create a plasma **314** in the plasma processing chamber **304** by providing an inductively coupled power. A pinnacle **372** extends from a chamber wall **376** of the plasma processing chamber **304** to the dielectric inductive power window **312** forming a pinnacle ring. The pinnacle **372** is angled with respect to the chamber wall **376** and the dielectric inductive power window **312**. For example, the interior angle between the pinnacle **372** and the chamber wall **376** and the interior angle between the pinnacle **372** and the dielectric inductive power window **312** may each be greater than 90° and less than 180°. The pinnacle **372** provides an angled ring near the top of the plasma processing chamber **304**, as shown. The TCP coil (upper power source) **310** may be configured to produce a uniform diffusion profile within the plasma processing chamber **304**. For example, the TCP coil **310** may be configured to generate a toroidal power distribution in the plasma **314**. The dielectric inductive power window **312** is provided to separate the TCP coil **310** from the plasma processing chamber **304** while allowing energy to pass from the TCP coil **310** to the plasma processing chamber **304**. A wafer bias voltage power supply **316** tuned by a bias matching network **318** provides power to an electrode **320** to set the bias voltage when a stack is placed on the electrode **320**. A cover **366** is placed over the electrode **320**. In this embodiment, the cover **366** is a bare silicon wafer. A controller **324** controls the plasma power supply **306** and the wafer bias voltage power supply **316**.

(16) The plasma power supply **306** and the wafer bias voltage power supply **316** may be configured to operate at specific radio frequencies such as, for example, 13.56 megahertz (MHz), 27 MHz, 2 MHz, 60 MHz, 400 kilohertz (kHz), 2.54 gigahertz (GHz), or combinations thereof. Plasma power supply **306** and wafer bias voltage power supply **316** may be appropriately sized to supply a range of powers in order to achieve the desired process performance. For example, in one embodiment, the plasma power supply **306** may supply the power in a range of 50 to 5000 Watts, and the wafer bias voltage power supply **316** may supply a bias voltage in a range of 20 to 2000 volts (V). In addition, the TCP coil **310** and/or the electrode **320** may be comprised of two or more sub-coils or sub-electrodes. The sub-coils or sub-electrodes may be powered by a single power supply or powered by multiple power supplies.

(17) As shown in FIG. 3, the processing chamber system **300** further includes a gas source/gas supply mechanism **330**. The gas source **330** is in fluid connection with plasma processing chamber **304** through a gas inlet, such as a gas injector **340**. The gas injector **340** may be located in any advantageous location in the plasma processing chamber **304** and may take any form for injecting gas. Preferably, however, the gas inlet may be configured to produce a “tunable” gas injection profile. The tunable gas injection profile allows independent adjustment of the respective flow of the gases to multiple zones in the plasma process chamber **304**. More preferably, the gas injector is mounted to the dielectric inductive power window **312**. The gas injector may be mounted on, mounted in, or form part of the power window. The process gases and by-products are removed from the plasma process chamber **304** via a pressure control valve **342** and a pump **344**. The pressure control valve **342** and pump **344** also serve to maintain a particular pressure within the plasma processing chamber **304**. The pressure control valve **342** can maintain a pressure of less than 1 torr during processing. An edge ring **360** is placed around a top part of the electrode **320**. The gas source/gas supply mechanism **330** is controlled by the controller **324**. For example, Kiyo® tools made by Lam Research Corp. of Fremont, CA, may be used to practice an embodiment.

(18) When the performance of the component is sufficiently degraded or after a specified number of wafers is processed or after a specified time of usage passes, the component is removed from the chamber for cleaning (step **108**). For example, the component may be removed after **300** radio frequency (RF) hours. RF hours is the amount of time that RF power is applied in the plasma chamber. FIG. 4A is an enlarged cross-sectional view of part of a component of the plasma processing chamber **304**, shown in FIG. 3, such as the inductive power window **312** after the

component has been removed. In this embodiment, processing the stacks deposits a residue layer **408** on plasma facing surfaces of the component. In this embodiment, the residue layer **408** comprises residues of tantalum, beryllium, platinum, manganese, cobalt, iron, ruthenium, magnesium, titanium, and silicon oxide. Of these residues, ruthenium and platinum are noble metals. In some embodiments, the residue layer **408** may be formed by thin layers of residues of different materials.

(19) After the component is removed from the plasma processing chamber **304** a Ru clean process is provided (step **112**). In an embodiment, the component is cleaned using at least one of hypochlorite and ozone (O.sub.3). In various embodiments, the hypochlorite may be at least one of sodium hypochlorite (NaClO), ammonium hypochlorite (NH.sub.4ClO), hydrogen hypochlorite (HClO). In some embodiments, the clean process may be a wet clean process, such as using a wet clean bath. For example, the wet clean bath may be an aqueous solution containing at least one of hypochlorite and ozone. In addition, the wet clean bath may further comprise ammonia. Although the Ru is described in this example, the composition may alternatively or additionally be used to clean residues containing other materials. In some embodiments, the cleaning may be performed while the component is in the chamber.

(20) FIG. 5 is a schematic view of the component **312** in an exemplary Ru wet clean bath **500**. The Ru wet clean bath comprises a Ru wet clean solution **504** comprising an aqueous solution of at least one of hypochlorite and ozone (O.sub.3) with ammonia in a container **508**. The residue layer **408** is exposed to the Ru wet clean solution **504**. In this embodiment, transducers **512** are provided. The transducers **512** provide ultrasonic or megasonic energy in order to increase residue removal. The Ru wet clean bath **500** removes at least some of the Ru residue. The component is removed from the Ru wet clean bath **500**. The component **312** is removed from the Ru wet clean bath **500**.

(21) A noble metal clean is provided (step **116**). In an embodiment, the component is cleaned using a noble metal cleaning composition comprising a mixture of thionyl chloride and pyridine. In some embodiments, the clean process may be a wet clean process, such as using a wet clean bath. For example, the wet clean bath may be an aqueous solution containing thionyl chloride and pyridine in a solvent. The solution of a mixture of thionyl chloride and pyridine in a solvent is known as organic aqua regia. Organic aqua regia is known for dissolving some noble metals, such as palladium, gold, and platinum. However, organic aqua regia does not dissolve ruthenium, rhodium, iridium, and osmium. In some embodiments, the solvent may be at least one of acetonitrile (CH.sub.3CN), dimethyl sulfoxide (DMSO), and dimethylformamide (DMF). Examples of the noble metal wet clean bath may be similar to the Ru wet clean bath **500**, with the difference being that the noble metal wet clean bath contains a noble metal wet clean solution of a mixture of thionyl chloride and pyridine in a solvent, such as water. The residue layer **408** is exposed to the noble metal cleaning composition. The noble metal cleaning composition removes Pd and Pt containing residues. The component **312** is removed from the noble metal cleaning composition.

(22) A silicon oxide clean is provided (step **120**). In an embodiment, the component is placed in a silicon oxide wet clean bath comprising an acid and a passivating solution. In this example, the acid may be at least one of hydrofluoric acid (HF) and ammonium fluoride (NH.sub.4F). The passivating solution may generally be an organic solvent chosen from at least one of the group of amines, alcohol, glycol, and acetone. For example, the passivating solution may be at least one of methylamine, ethyleneamine, ethylenediamine, diethylenetriamine, ethylene glycol, propylene glycol, and acetylacetone. The acid removes silicon oxide containing residue. The passivating solution protects the surface of the component. The silicon oxide wet clean bath may be similar to the Ru wet clean bath **500**, with the difference being that the silicon oxide wet clean bath contains a silicon oxide wet clean solution. The residue layer **408** is exposed to the silicon oxide wet clean solution. The silicon oxide wet clean removes silicon oxide containing residues. The passivating solution passivates the surface of the component and prevents damaging of the texture applied on the parts. The component is removed from the silicon oxide wet clean bath.

(23) In this embodiment, silicon oxide wet clean bath further comprises other acids to remove other metals. The other acid may be one or more of hydrochloric acid (HCl), nitric acid (HNO₃), and sulfuric acid (H₂SO₄). Such acids may be used to remove one or more of Ta, Ni, Co, Cr, Hf and III-V metals. In other embodiments, a separate bath of one or more of HCl, HNO₃, and H₂SO₄ may be used to remove one or more of Ta, Ni, Co, Cr, Hf and III-V metals before providing the silicon oxide wet clean bath.

(24) The steps of providing a Ru wet clean (step **112**), providing a noble metal wet clean (step **116**), and providing a silicon oxide wet clean (step **120**) may be repeated a plurality of times for a plurality of cycles (step **124**). The residue layer **408** may comprise alternating layers of different residues. The alternating layers may be caused by the different wafers being sequentially etched for many different wafers. FIG. **4B** is an enlarged cross-sectional view of part of a component **312** after the residue layer **408**, shown in FIG. **4A**, has been removed.

(25) The component **312** is then mounted in a plasma processing chamber (step **128**). The plasma processing chamber is used to process metal containing layers, such as etching metal containing layers (step **132**).

(26) The ability to clean metal containing residues from the component, allows for the component **312** to be reused, instead of requiring disposal of the component. Reuse of the component **312** reduces costs and waste, reduces contamination of devices being processed, and increases device uniformity. Various embodiments use organic aqua regia instead of regular aqua regia in order to reduce damage to the component surface.

(27) In some embodiments, additional cleaning steps may be provided. For example, a deionized water rinse may be provided after any of the wet cleaning steps. In other embodiments, additional wet cleaning steps may be provided. Other cleaning processes may be added in other embodiments. For example, scrubbing or beam blasting may be provided in other embodiments.

(28) In other embodiments, other steps may be provided after the component **312** is again mounted in a plasma processing chamber (step **128**) and before using the component in the plasma processing chamber (step **132**). For example, a plasma process may be used to condition or season the component **312**.

(29) In other embodiments, metal layers, such as a Ru containing layer may be used as a hardmask. Such embodiments may not have a step of etching a metal containing layer. However, residue from the Ru containing layer may be deposited on parts of the plasma processing chamber. Other embodiments may have other metal containing layers for other uses.

(30) While this disclosure has been described in terms of several preferred embodiments, there are alterations, permutations, modifications, and various substitute equivalents, which fall within the scope of this disclosure. It should also be noted that there are many alternative ways of implementing the methods and apparatuses of the present disclosure. It is therefore intended that the following appended claims be interpreted as including all such alterations, permutations, and various substitute equivalents as fall within the true spirit and scope of the present disclosure.

Claims

1. A method of cleaning residue comprising ruthenium (Ru) residue, noble metal residue, and silicon oxide residue on at least one surface of a component of a semiconductor processing chamber, comprising: exposing the residue to a Ru cleaning composition comprising at least one of hypochlorite and O₃ based chemistries, wherein the Ru cleaning composition removes the Ru residue; followed by exposing the residue to a noble metal cleaning composition comprising thionyl chloride and pyridine in a solvent, wherein the noble metal cleaning composition removes the noble metal residue; and followed by exposing the residue to a silicon oxide wet clean solution comprising an acid and a passivating solution, wherein the acid removes the silicon oxide residue and wherein the passivating solution passivates at least one surface of the component.

2. The method, as recited in claim 1, the exposing the residue to the Ru cleaning composition comprises immersing the component in a bath of the Ru cleaning composition.
 3. The method, as recited in claim 1, wherein the Ru cleaning composition further comprises ammonia.
 4. The method, as recited in claim 1, wherein the solvent is an organic solvent.
 5. The method, as recited in claim 4, wherein the organic solvent is at least one of dimethyl sulfoxide and dimethylformamide.
 6. The method, as recited in claim 1, wherein the exposing the residue to the Ru cleaning composition, the exposing the residue to the noble metal cleaning composition, and the exposing the residue to the silicon oxide wet clean solution are performed sequentially for a plurality of cycles.
 7. The method, as recited in claim 1, wherein the acid of the silicon oxide wet clean solution, comprises at least one of hydrofluoric acid (HF) and ammonium fluoride (NH₄F).
 8. The method, as recited in claim 7, wherein the passivating solution comprises at least one of methylamine, ethyleneamine, ethylenediamine, diethylenetriamine, and acetylacetone.
 9. The method, as recited in claim 7, wherein the passivating solution comprises an organic solvent.
 10. The method, as recited in claim 7, wherein the passivating solution comprises at least one of a group of amines and acetone.
 11. The method, as recited in claim 1, wherein the Ru cleaning composition further comprises water.
-